

# **ECON 002:**

## **Principles of Macroeconomics**

### **Lecture 7: Full Employment, Midterm Review**



# Types of Unemployment



3. **Structural Unemployment:** mismatch between skills of workers and needs of employers.

- Example: In 1900s, many jobs at AT&T, US West, BellSouth (Baby Bells).  
Cellular service and internet = fewer operators and jobs at landline companies.  
New jobs at? Internet service providers, Verizon, T-Mobile, Apple, Google
- Unemployment was generated by progress in the economy.  
Long-term problem for some people, not indicative of a poor economy.

4. **Cyclical Unemployment:** Job loss that results directly from changes in economic production.

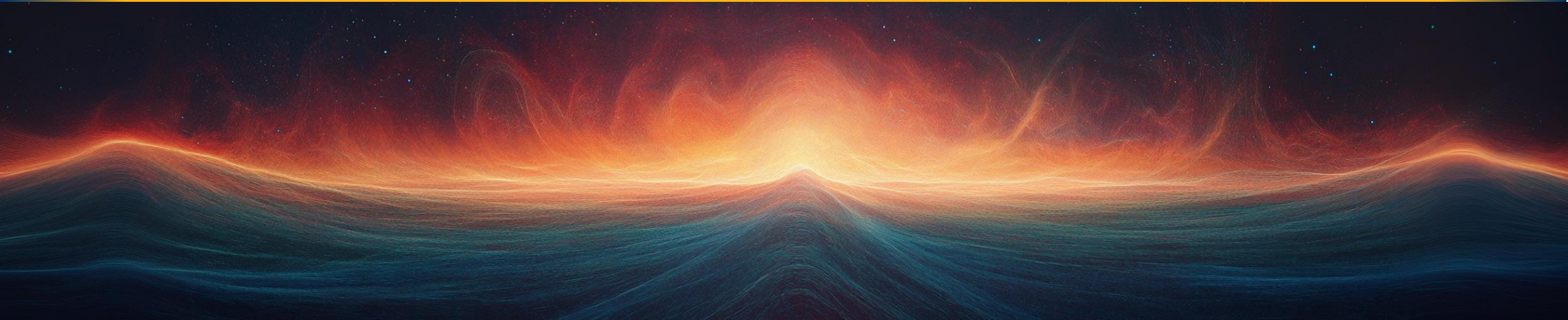
- Example: US Car Industry in 2009.  
High gas prices+gas-guzzling cars+housing market crash=permanent job loss.
- Cyclical unemployment is representative of problem in the entire economy.



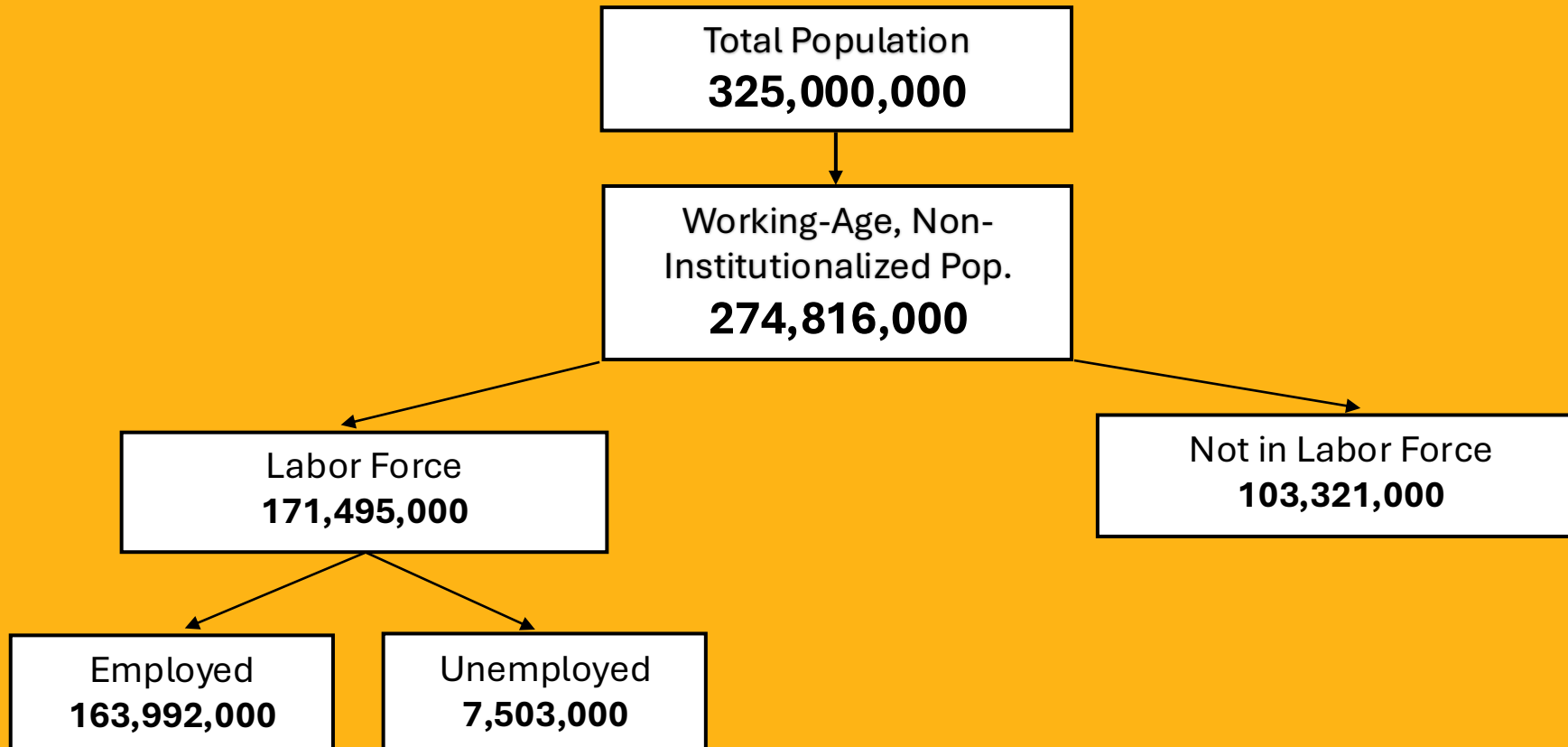


# Calculating the Unemployment Rate

- Bureau of Labor Statistics (BLS)
- Today there are ~325 million people in the US, 7.35 billion in the world! Do not expect all of them to be employed.
- BLS calls 60,000 households in the 2<sup>nd</sup> week of the month → households chosen so they represent the population
- In December, 2025:
- Civilian, Working-Age, Non-institutionalized Population = **274,816,000**
  - Age and Institutionalization Restrictions
  - Other Restrictions?



# A Picture of Employment



# Calculating the Unemployment Rate

- Civilian, Working-Age, Non-institutional Population = **274,816,000**

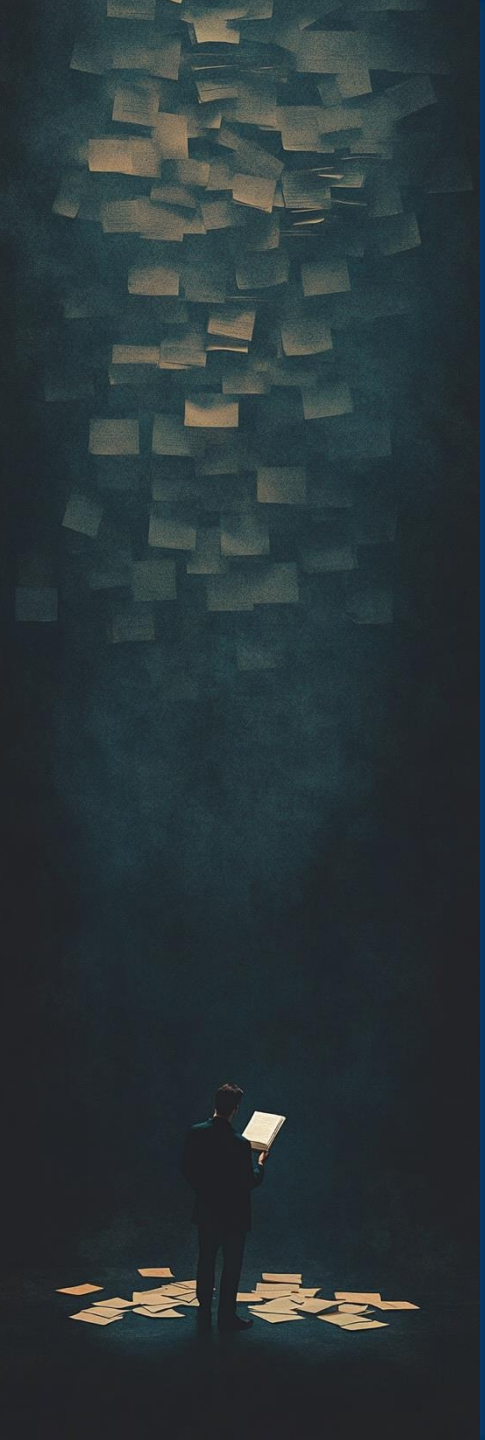
Employed Workers: any work done for pay in the past week

- part-time work
- temporary work
- didn't work last week because of vacation, illness, family, weather, labor dispute
- unpaid family work

- **163,992,000** employed workers in December, 2025

Unemployed Workers:

1. Did not work for at least one hour in the past week for pay.
  2. Actively searched for work in the last 4 weeks.
    - contacting employer, employment agency, submitted resumes, answering job ad
    - does not include passively looking for jobs online
- **7,503,000** unemployed workers in December, 2025



# Calculating the Unemployment Rate

- Unemployment Rate: the fraction of workers in the labor force who are unemployed

- $$\text{UE Rate} = \frac{\text{Number of unemployed}}{\text{Labor Force}} \times 100$$

- In December 2025, 
$$\text{UE Rate} = \frac{7,503,000}{171,495,000} \times 100 = 4.37\%$$

- Civilian Labor Force: Individuals who are employed or actively searching for work.

- $$\text{LF} = \text{Employed} + \text{Unemployed} = \mathbf{163,992,000 + 7,503,000 = 171,495,000}$$

- Labor Force Participation Rate = fraction of Working-Age, Non-institutional Population in the labor force.

- $$\text{Labor Force Participation} = \frac{\text{Number in LF}}{\text{Working-Age Pop.}} \times 100 = \frac{\mathbf{171,495,000}}{\mathbf{274,816,000}} \times 100 = 62.4\%$$

# Shortcomings of the UE Rate Calculation



- Two big problems
- 1.
- 2.
- Can we account for these shortcomings?



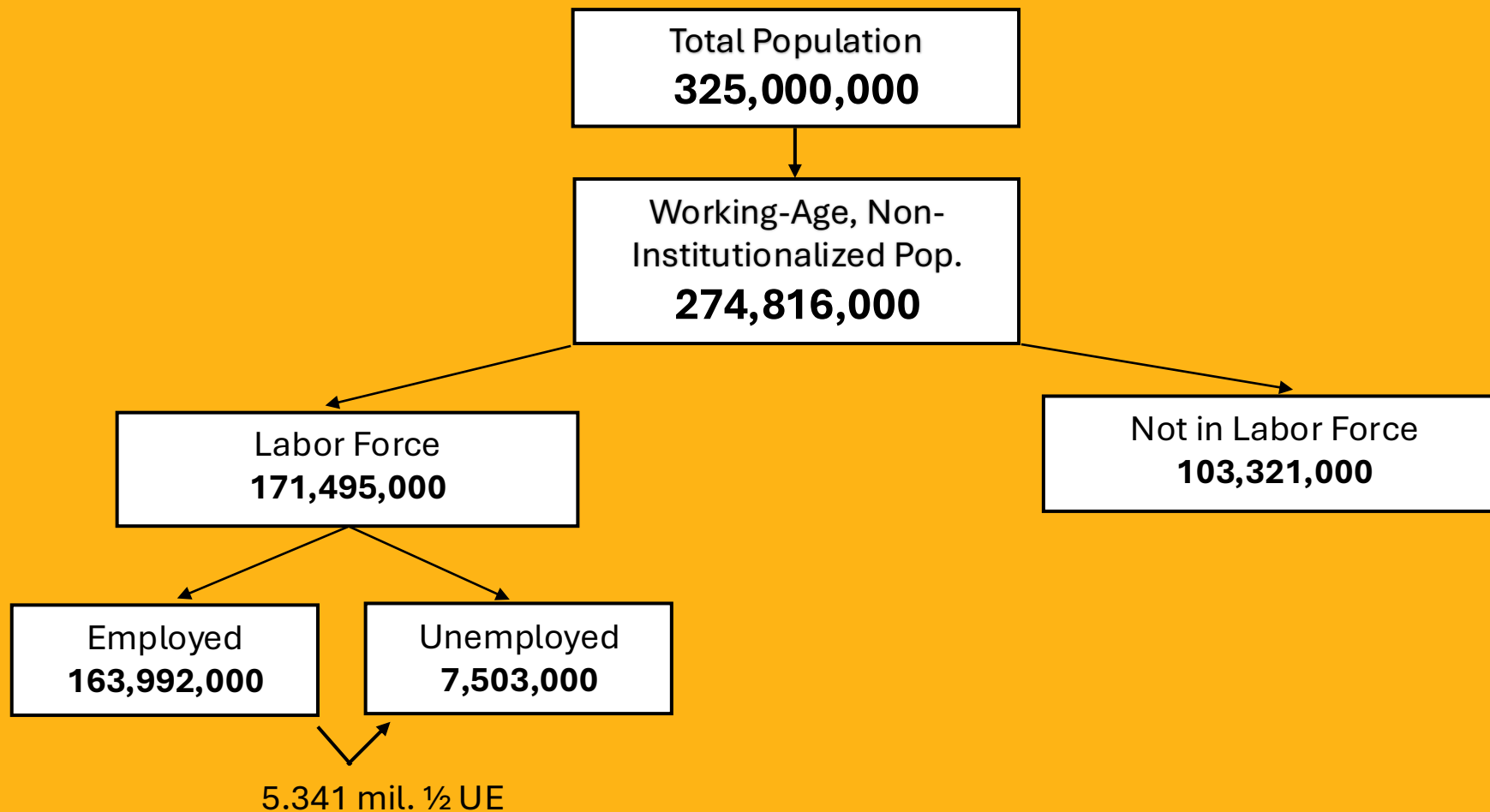
# Involuntary Part-Time Workers

- Many workers want a full-time job, but only have a part-time job.
- Example: Individual wants to work 40 hours a week, but can only find 20 hours of work.
  - This individual is  $\frac{1}{2}$  unemployed.
- Involuntary Part-Time Worker:
- Unemployment Rate with IPT Workers=





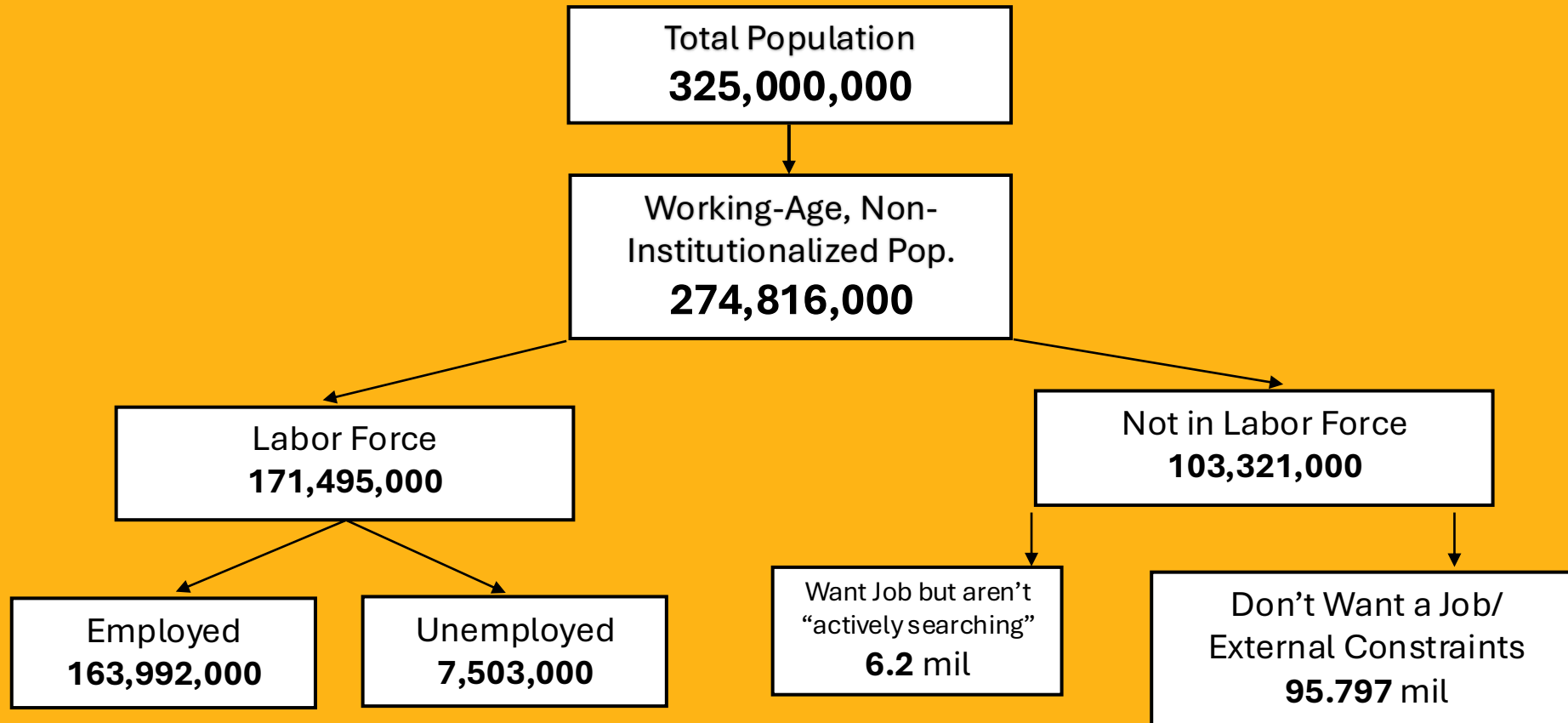
# A Picture of Employment



# A Picture of “Not in the Labor Force”

- Which workers would take a job if offered one?
- Discouraged Unemployment Rate =





# Unemployment Reporting

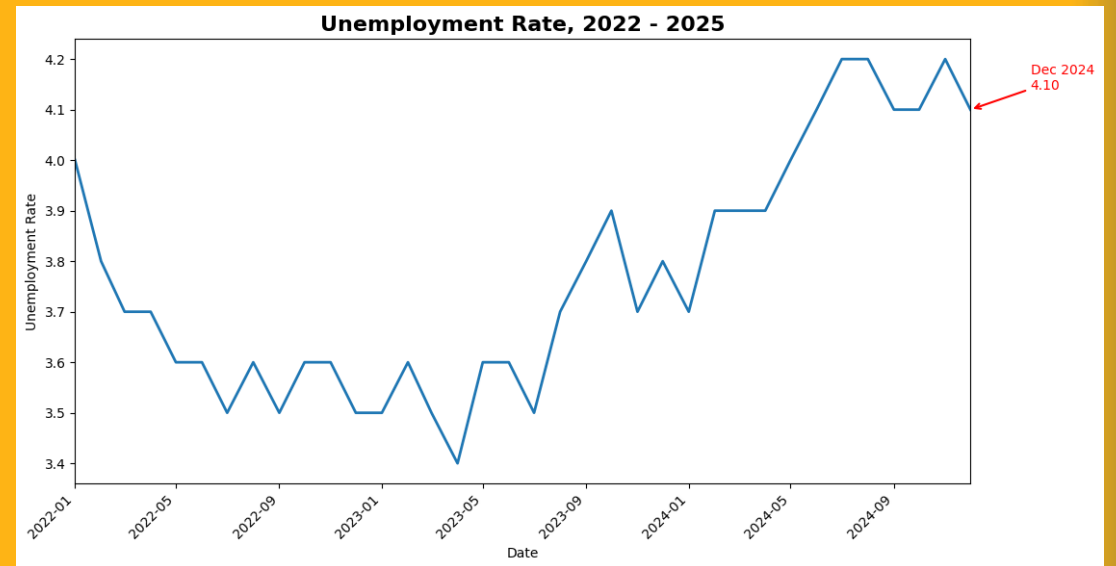
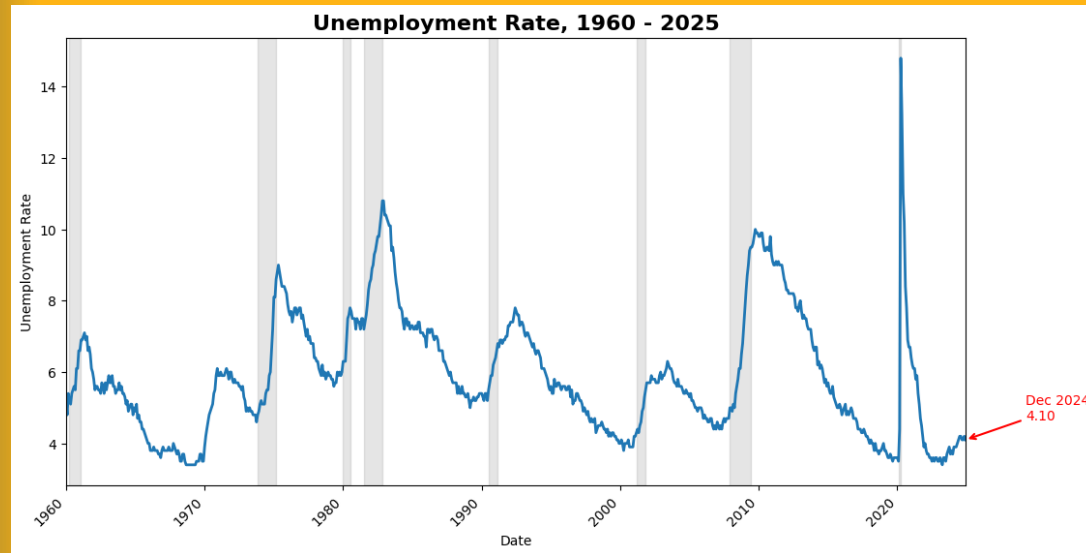
| Measure | Dec. 2023 (NSA) | Nov. 2024 (NSA) | Dec. 2024 (NSA) | Dec. 2023 (SA) | Aug. 2024 (SA) | Sept. 2024 (SA) | Oct. 2024 (SA) | Nov. 2024 (SA) | Dec. 2024 (SA) |
|---------|-----------------|-----------------|-----------------|----------------|----------------|-----------------|----------------|----------------|----------------|
| U-1     | 1.3             | 1.6             | 1.5             | 1.4            | 1.6            | 1.6             | 1.7            | 1.7            | 1.6            |
| U-2     | 1.8             | 1.9             | 1.9             | 1.8            | 2.0            | 1.9             | 2.0            | 2.0            | 1.9            |
| U-3     | 3.5             | 4.0             | 3.8             | 3.8            | 4.2            | 4.1             | 4.1            | 4.2            | 4.1            |
| U-4     | 3.7             | 4.2             | 4.1             | 4.0            | 4.4            | 4.3             | 4.4            | 4.5            | 4.4            |
| U-5     | 4.4             | 4.9             | 4.7             | 4.7            | 5.0            | 5.0             | 5.0            | 5.1            | 5.0            |
| U-6     | 7.0             | 7.4             | 7.4             | 7.2            | 7.8            | 7.7             | 7.7            | 7.7            | 7.5            |

## Notes:

- **U-1:** Long-term unemployment (15 weeks or longer).
- **U-2:** Job losers and persons who completed temporary jobs.
- **U-3:** Official unemployment rate.
- **U-4:** U-3 + discouraged workers.
- **U-5:** U-4 + marginally attached workers.
- **U-6:** U-5 + part-time workers for economic reasons.



# Unemployment Today



# Midterm 1 Notes

- Monday, February 2<sup>nd</sup> : ~9:30 – 10:45am
- 30 Multiple Choice Questions: ~2.5 minutes/question
- Evenly distributed among topics of
  - Supply/Demand (Chapter 2)
  - GDP (Chapter 3.1)
  - CPI (Chapter 3.2)
  - Unemployment (Chapter 3.3)
- Also due on Monday, February 2<sup>nd</sup> at 9:30am
  - Lecture Quiz 4

## Equations

$$\text{Growth Rate}_{YTA, YTB} = \frac{\text{Value}_{YTB} - \text{Value}_{YTA}}{\text{Value}_{YTA}} \times 100$$

$$\text{CPI}_{\text{Current Year}} = \frac{\text{Basket Cost in Current Year}}{\text{Basket Cost in Base Year}} \times 100$$

$$\text{Weighted Basket Cost}_{\text{Current Year}} = P_{\text{Current Year}}^{\text{Goods}} \times Q_{\text{Base Year}}^{\text{Goods}}$$

$$\text{GDP Deflator}_{\text{Current Year}} = \frac{\text{Nominal GDP in Current Year}}{\text{Real GDP in Current Year}} \times 100$$

$$\text{Unemployment Rate} = \frac{\# \text{ Unemployed}}{\# \text{ in Labor Force}} \times 100$$